

Enterprise Digital Infrastructure – Self-assessment– Email

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First name/Lastname _____

ID _____

- A. A user named Poldo opens his favorite MUA running on his device and sends three mail messages (one after the other) to the Mail Submission Agent running on his Local Mail Server. The device and the Local Mail Server belong to the domain **mail.me**.
- The first message has three recipients: **student1@eva.unipv.it**, **student1@poldo.unipv.it** and **student2@poldo.unipv.it**
 - The second message has two recipients: **info@mail.me** and **student3@eva.unipv.it**
 - The third message has one recipient: **andy@gmail.com**
1. Compute the number of lookup phases, the Name Servers involved and the number of RTTs for each of lookup phase.
 2. Compute the number of TCP connections opened by the MUA.
 3. Does Poldo need to authenticate with the Mail Server before sending the messages? Why?
 4. Compute the number of RTTs for sending the information associated with the envelopes of the three messages.
- B. The messages received by the Mail Submission Agent are processed by the Mail Transfer Agent running on the Local Mail Server for their delivery.
1. Compute the number of lookup phases, the Name Servers involved and the number of RTTs for each of lookup phase.

2. Compute the number of TCP connections opened by the MTA.
3. Compute the number of RTTs for sending the information associated with the message envelopes.
4. Under the assumption that the Mail Server of the domain **mail.me** implements the SPF, DKIM and DMARC technologies, compute the number of queries necessary to validate/authenticate the messages.